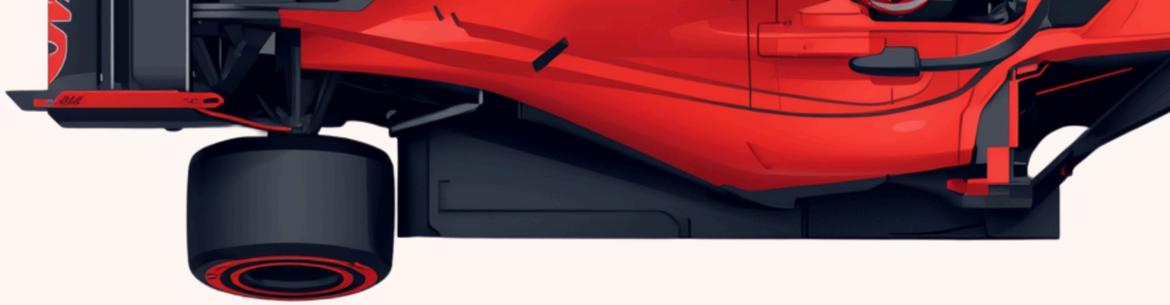




ME5670: Vehicle Dynamics and Modelling

Optimal Autonomous Control of an F1 Car with Active Aerodynamics: A Nonlinear MPC Approach

"I knew he would brake because he has a wife and two kids at home." - **Rookie Alonso**

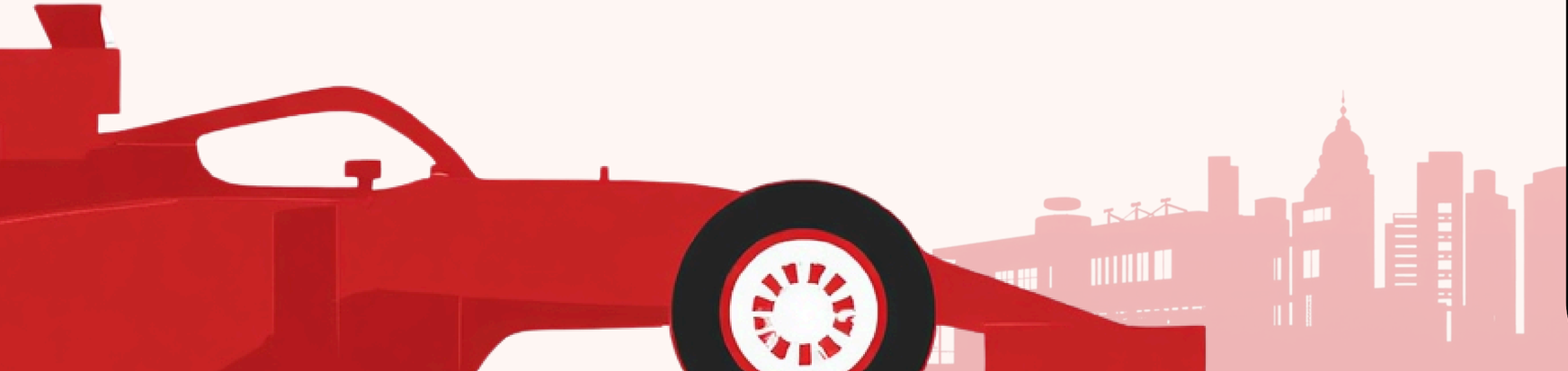


Team XTorque

- Ashish Sam
- Frazel Fernandes
- Joel J Nellikkunnel
- Mukhil M
- Sanjai V

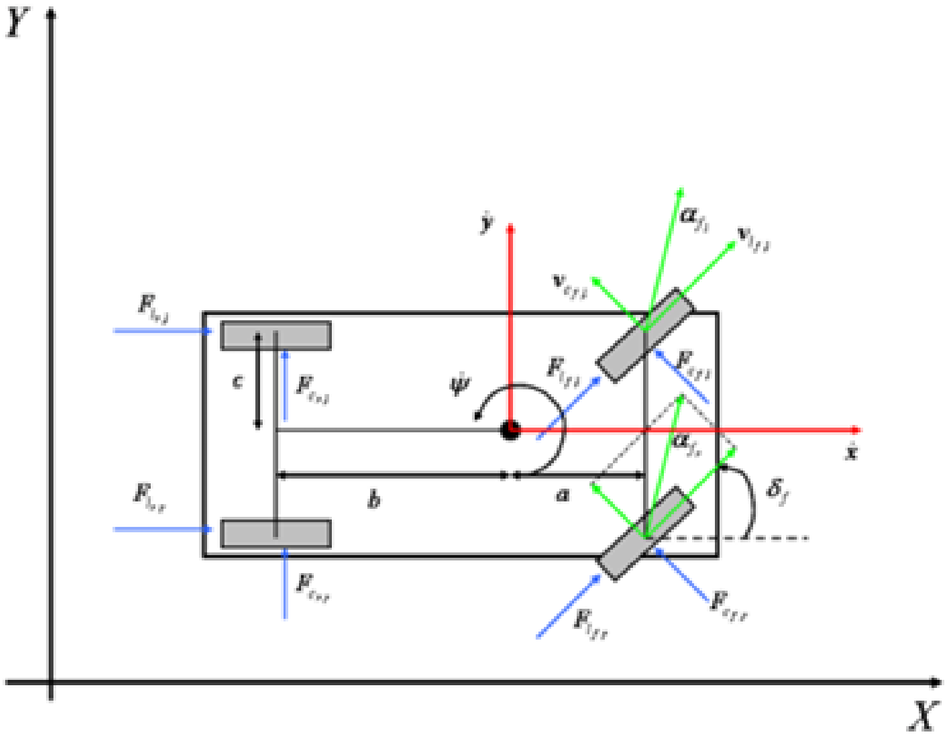
Problem Formulation and the 130R challenge

- The objective of this study is to formulate an NMPC to autonomously navigate the high-speed 130R corner at Suzuka Circuit. The vehicle plant features continuous active aerodynamics, requiring the controller to simultaneously optimize steering, throttle, and the front and rear wing Angles of Attack (AoA).
- The primary challenge lies in operating at the absolute limits of tire friction, where the solver must autonomously balance aerodynamic drag on the straights with the massive downforce generation required to survive the curve.





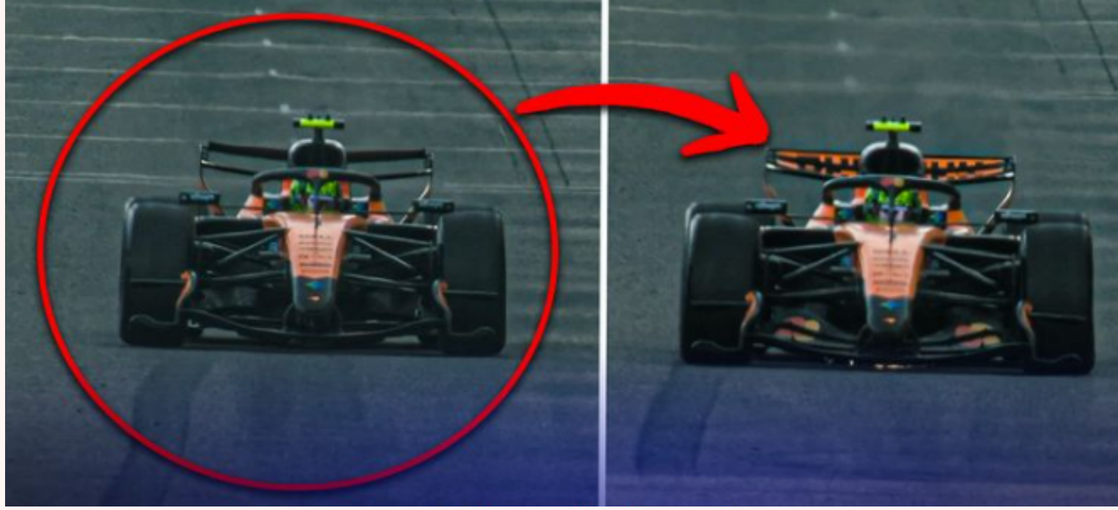
Vehicle and Tire Dynamics



Credits: Falcone et al., 2007

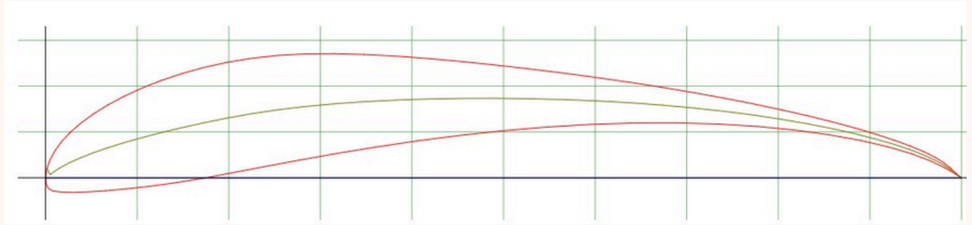
- Vehicle Dynamics is formulated through a 7DOF double-track model, considering the downforces generated by the wings, as well as induced drag, parasitic drag, and rolling resistance.
- The vehicle is considered to be a RWD with all-wheel braking, with a brake bias of 0.58 at the front.
- Load transfer during cornering is considered to determine the dynamic normal loads acting on the tire.
- A combined Pacejka model is used to determine the simultaneous lateral and longitudinal forces acting on the tire.

Active Aero Integration



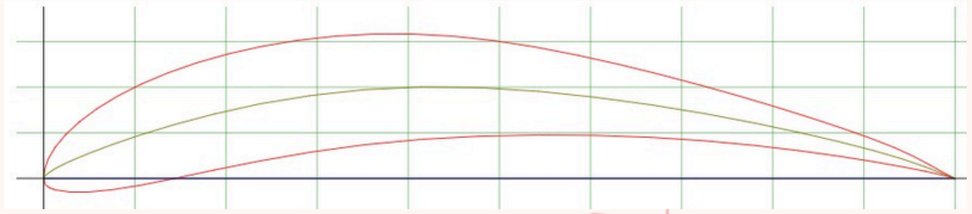
Active aero- a new feature of the 2026 F1 Regulations
Wings are actively controlled, not fixed like conventional cars

F1 2026 cars : 2 configurations (X-mode and Z-mode)
Our vehicle : Real-time continuous control
Front and rear wings act as control inputs



Front Wing – Selig S1223

- Heavily cambered; high-lift at low Reynolds numbers
- Popular among FSAE teams



Rear Wing – Eppler E423

- Lower downforce but gentler stall characteristics
- Predictable lift curve



130R Track Model

About 130R Corner (Suzuka Circuit)

- High-speed left-hand turn (~130 m radius)
- Generates up to 5G lateral forces
- Requires precise control & optimal racing line

Key Challenges

- Extreme lateral grip limits
- Weight transfer to right-side tires
- Maintaining stability at maximum speed

Actual 130R Curve: Double-apex corner

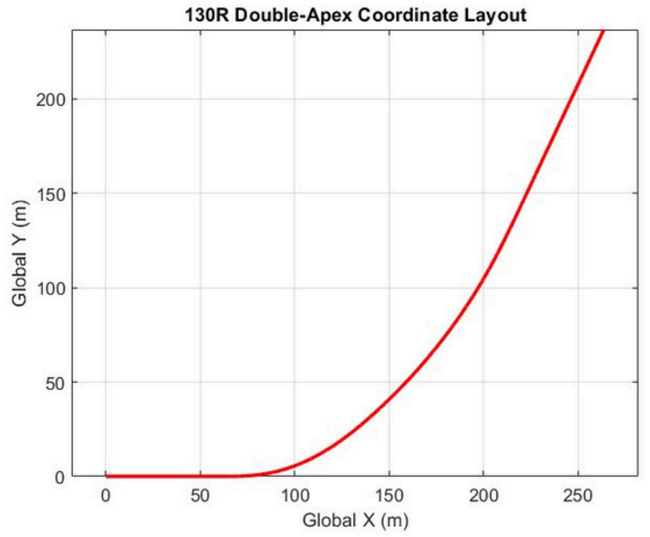
- First apex radius: 85m
- Second apex radius: 340m

Modeling Approach

- Built using curvature (κ) profile
- Uses straights + clothoids + arcs
- Integrated to get (X, Y, ψ)
- Smooth, continuous







Track Mapping: The Frenet–Serret Frame

Why not standard MPC reference formulation:

- Constraints in (X, Y, ψ, V_x_ref)
- Track limits become nonlinear & complex
- Pre-made formulation; penalizes a faster approach

Frenet Frame Idea

- Transform global coordinates → track-aligned frame
- Defined along centerline
- s: Arc length
- e_y: Lateral error
- e_psi: Heading error

$$e_\psi = \psi - \psi_{ref}(s)$$
$$\dot{s} = \frac{V_x \cos(e_\psi) - V_y \sin(e_\psi)}{1 - \kappa(s)\, e_y}$$
$$\dot{e}_y = V_x \sin(e_\psi) + V_y \cos(e_\psi)$$

How it helps

- Track limits → $-w/2 \leqslant ey \leqslant w/2$
- No need for Vref
- Cleaner, faster optimization

NMPC Strategy and Cost Framework

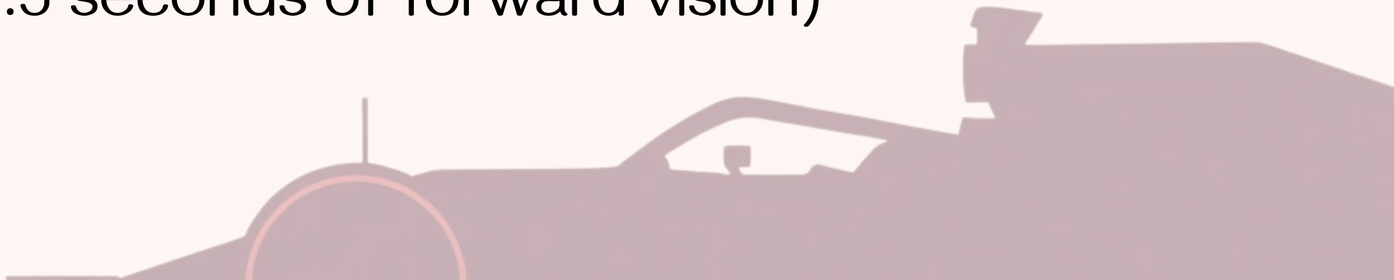
Cost Function Design :

The NMPC minimizes lap time through the corner by optimizing:

$$J = \text{Rewards} + \text{Tracking Penalties} + \text{Control Smoothing}$$

Where the objective balances:

- Maximizing forward spatial progression (S) and velocity - Rewards
- Minimizing lateral deviation (e_y) from the racing line - Tracking Penalties
- Penalizing aggressive actuator slew rates (Delta u) for smoothness - Control smoothing
- Prediction Horizon: N = 50 steps
- Sampling Time: Ts = 0.05s (yielding 2.5 seconds of forward vision)



Constraints

Constraints & Real-Time Optimization

State Constraints:

- Tire Dynamics: Pacejka Magic Formula limits with load-sensitive grip
- Track Boundaries: Strict lateral position limits
- Powertrain Limit: Wheel torque bounded by 750 kW max power

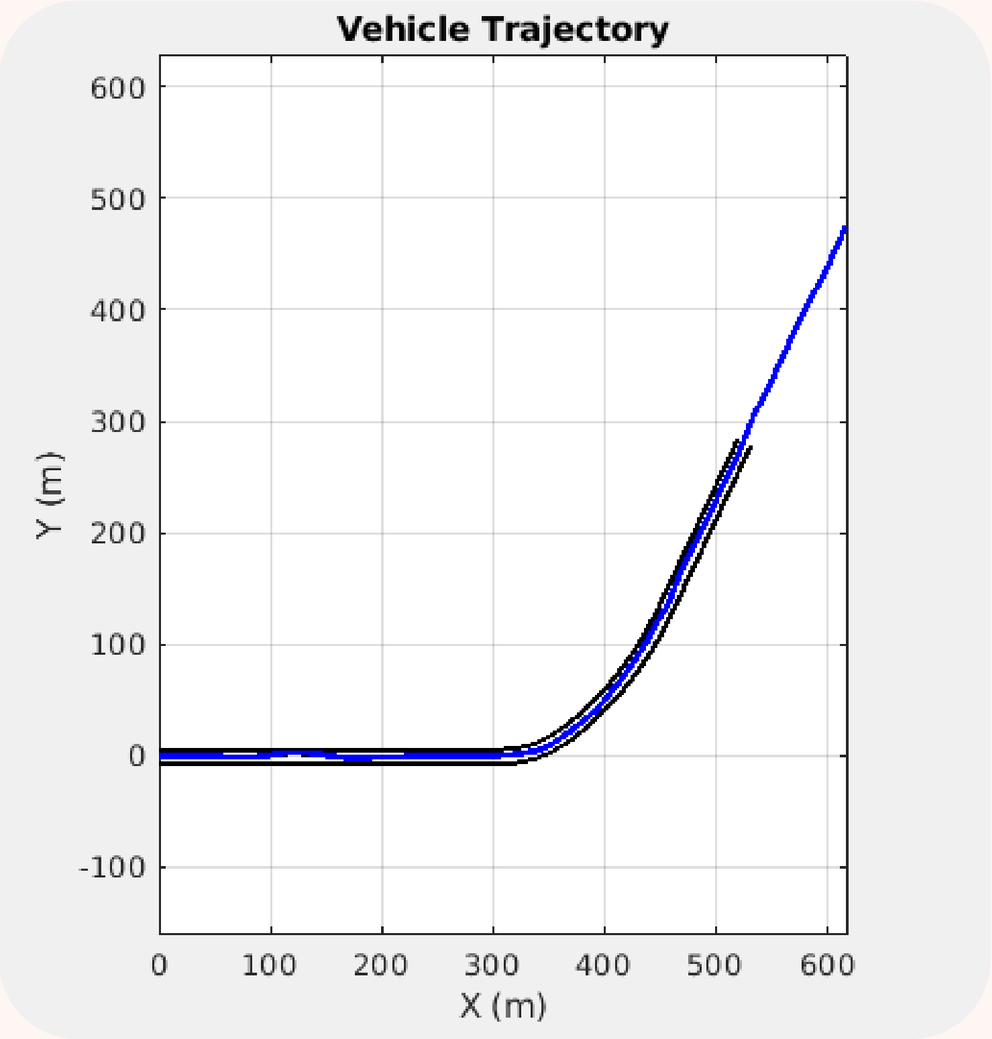
Control Constraints:

- Unified Torque: $T_{\text{brake}} \leq T \leq T_{\text{drive}} \quad (\text{Nm})$
- Steering Angle: ± 30° continuous limit
- Active Aero: Continuous Angle of Attack limits for Selig S1223 (Front) and Eppler E423 (Rear) airfoils

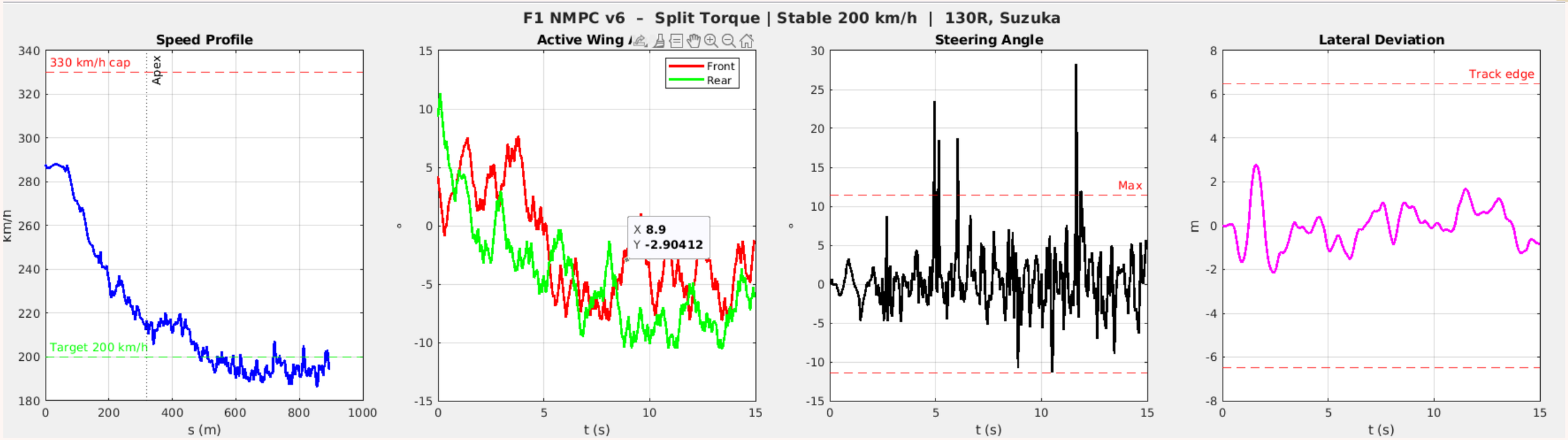
Solver Architecture:

- Toolchain: CasADi compiled with IPOPT
- Real-time feasibility: Achieved via constraint softening and shifting warm-starts.

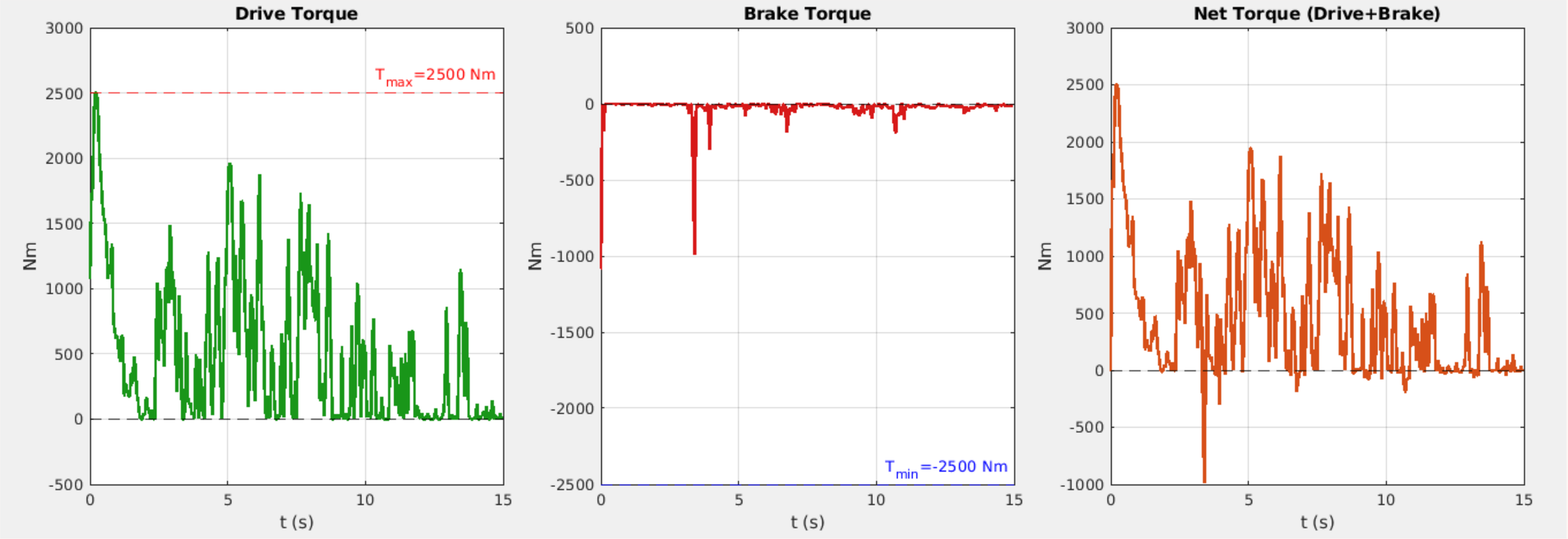
Simulation Results



Simulation Results

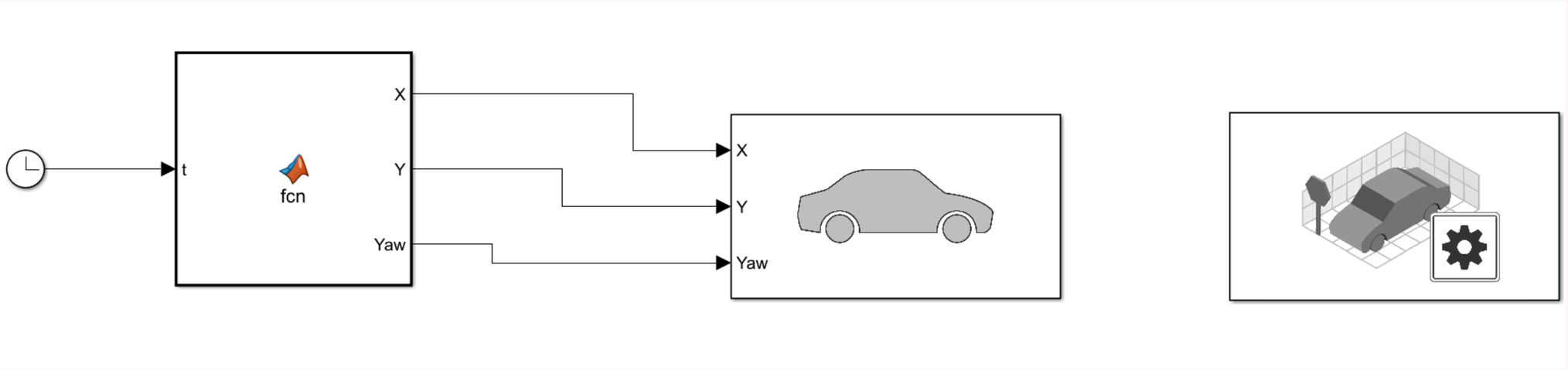


Simulation Results

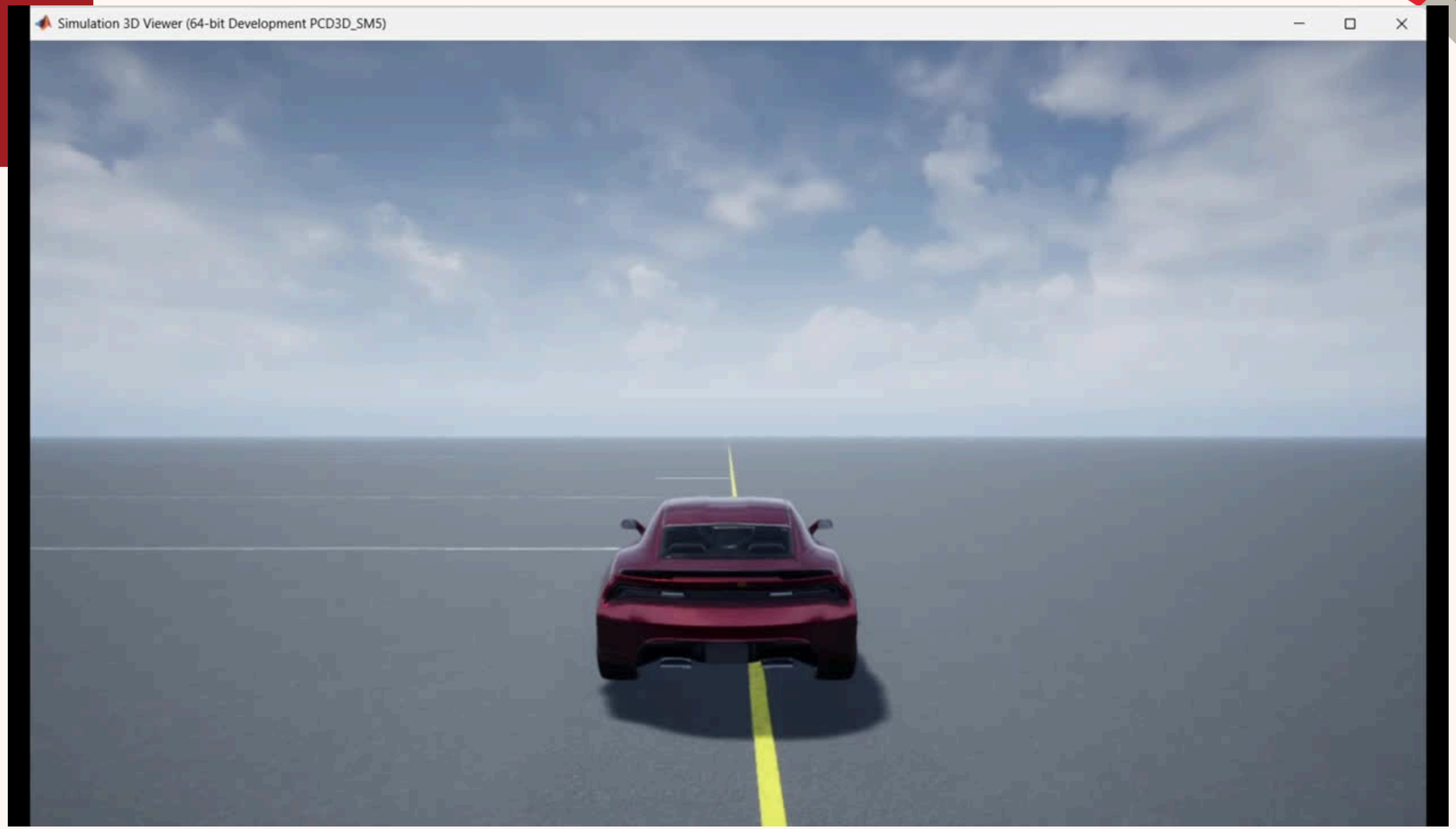


Unreal Engine–Simulink Integration

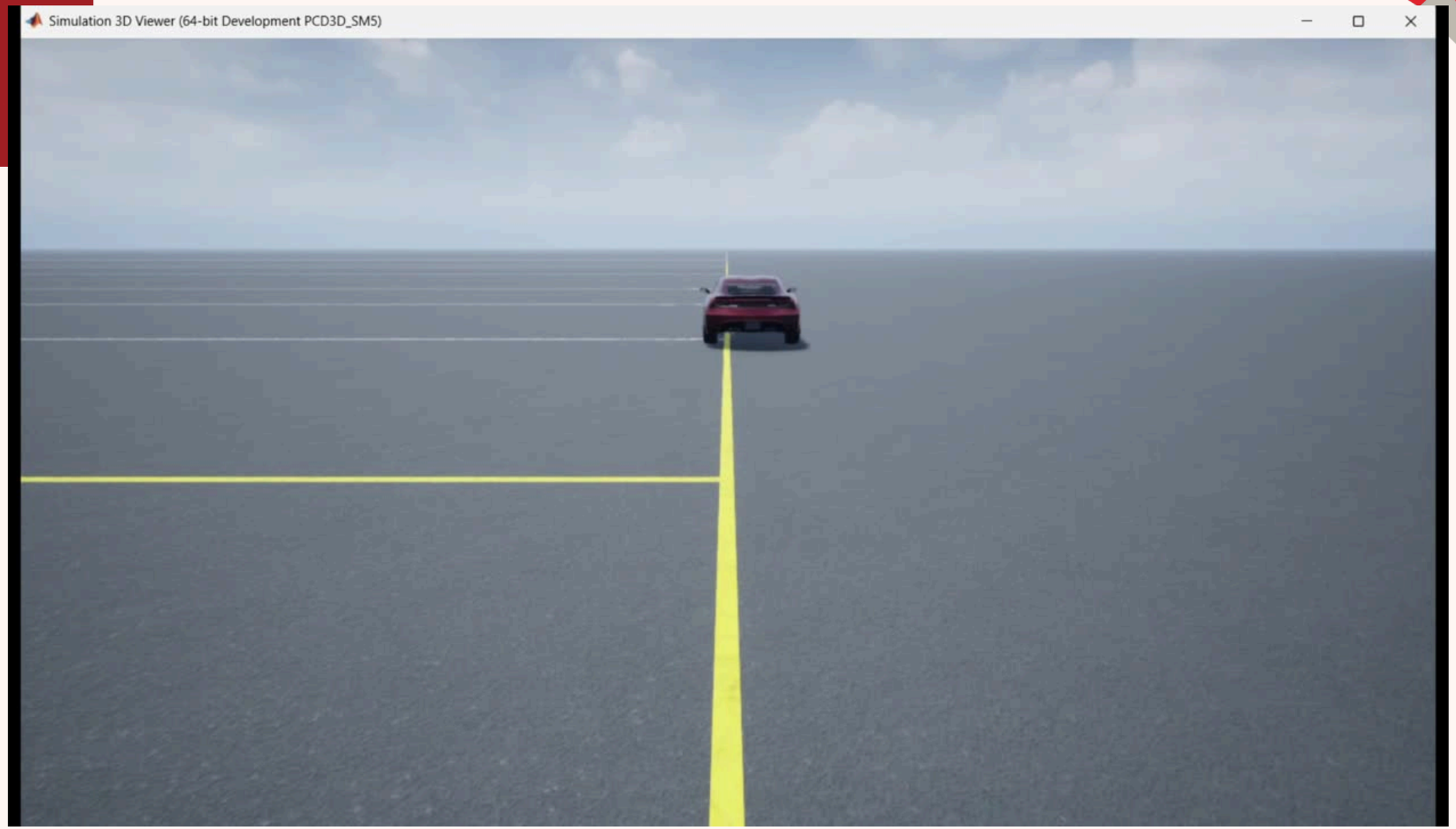
- Simulink is used to develop an animation of the car the Unreal Engine Visualizer using a default car and environment.
- A **MATLAB function block** is used to create a custom script that loads the data from the simulation and outputs the X, Y and yaw data of the vehicle at every instant.
- A **Simulation 3D Vehicle with Ground Following** block is used to simulate the motion of the vehicle.

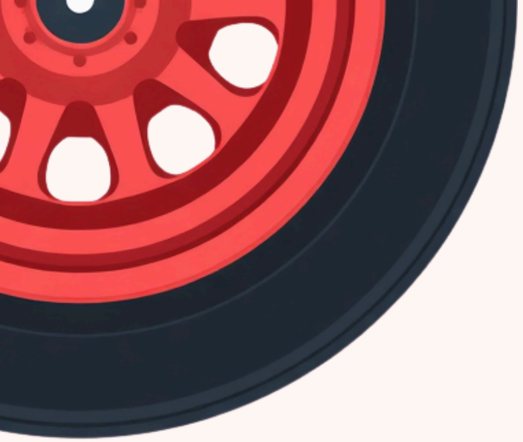


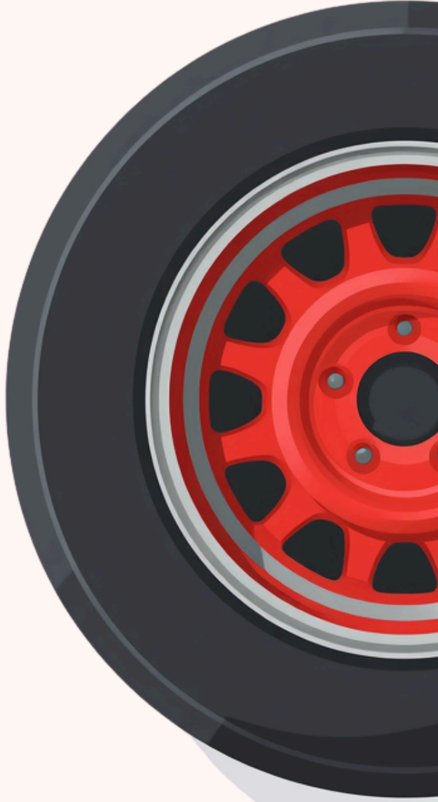
Unreal Engine–Simulink Integration



Unreal Engine–Simulink Integration







Thank You!

